Sprint Planning

Sprint #1

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Introduction

The objective of this meeting is to plan the first Sprint of your project. The final result of the following tasks should be the Sprint Backlog ready for the first Sprint.

Architecture of created systems

In order to maintain the consistency of projects and to enable the implementation of all design tasks, all created systems will have a very similar architectural frame. Its outline was presented in the lecture on Software Architecture. The frame itself in the form of the base design will also be available to each team.

Assign team members

Join your team via the eKursy form.

Project assignment

First, the laboratory teacher will assign projects to teams.

Choosing roles in the team

From the team members, you are to choose two persons for the following roles:

- **Scrum Master** – a person who will be responsible for adhering to the principles of the Scrum methodology and increasing the productivity of the team. This person will also carry out programming tasks.
- **Product Owner Proxy** – the person performing this role will maintain the Product Backlog (or clarifying the requirements at the request of the Product Owner). In the following Sprint(s), PO proxy may be asked to propose new Product Backlog items and take over the Product Owner role.

The role of the **Product Owner** will be performed by the laboratory teacher.

Product Backlog (PB)

Download a spreadsheet from Moodle containing the description, Product Backlog and starting point for the Sprint Backlog of the first Sprint.
Finally, the information from this spreadsheet will be entered into *Trello*, *GitHub* (or another tool) for managing

tasks, but at this stage we will work with the worksheet to start planning the first Sprint as quickly as possible. To facilitate communication within team, it is best if you put this sheet "in the cloud" (eg Google Docs).
The sheet contains the following tabs:

- **Project description** – name and a short description of the project to be implemented (in some cases there is also some additional description, e.g. data format)
- **Product Backlog** – prepared and pre-arranged according to importance by the Product Owner
- **Sprint Backlog** – A working area to enter the Sprint Backlog for the first Sprint. It contains the Sprint Goal, the Product Backlog items you choose to include in the Sprint scope, and the tasks for each of these items. Any other tasks you need to complete in the first Sprint should also be listed here.

Definition of Done and other scoring requirements

Before we start planning, we should read the Definition of Done (DoD) and other project scoring requirements. There is a link on Moodle called "Sprint Grading Rules". This document contains the requirements and scores for individual elements of the Definition of Done and other requirements related to the development of the project.
Familiarise yourself with the rules and decide in your team which elements of the DoD you will want to implement. This will allow you to better plan your work.
In the future, it is worth placing a link to this document in the project's README.md file or in another place where it will be easy and quick to access.

"Capacity" estimation

In order to be able to plan a sprint, we still need to estimate the team's hourly capacity.
Sprint #1 deadline:

- Monday groups: 18-05-2026 (Erasmus:25-05-2026)
- Tuesday group: 19-05-2026

Take a look at the calendar. Let each of you estimate how many hours you will be able to spend on the project in the sprint you are starting today.

Planning the scope of the Sprint

At this stage, we need a Product Owner. Report your readiness to the laboratory teacher. While waiting, take a look at the items of the Product Backlog and the requirements contained in the DoD.
The Product Owner will explain the details related to the items of the Product Backlog that he would like to implement in the next Sprint. Try to define a Sprint Goal together.
When the teacher moves to other teams, your task will be to estimate the items of the Product Backlog (at least the number of hours of the estimated availability of the team).

Product Backlog items estimation

Estimate the labour intensity of individual PB items, e.g. by playing Planning Poker (you can use online tools, e.g. https://www.planitpoker.com).
Complete the column of estimated labour intensity in the spreadsheet in the Product Backlog.
When estimating and planning, it is important to take into account the fact that the gain must comply with the Definition of Done (and also take into account your uncertainty at this stage in relation to what is included in it).

Product Backlog items selection

Ask the Product Owner again. Go to the Product Backlog tab and together select the PB items to be implemented as part of the Sprint, taking into account the team's time availability and estimates. Copy the selected Product Backlog items to the Sprint Backlog tab.

Decomposition into tasks

The next step is to create tasks in the Sprint Backlog. The Sprint Backlog should also include tasks (or more broadly speaking, a plan) which do not result directly from specific items of the Product Backlog, but you will have to do them (e.g. configuring *Github*, *Travis*, *Trello*, uploading the project to *Github*, obtaining general knowledge about technology, etc.).
If it turns out that after breaking down the Product Backlog items into tasks, their total labour intensity exceeds the previous estimates, you should ask the Product Owner and "renegotiate" the scope.
At this point, you just need to define the time and place of the Daily Scrum (it should be carried out on all days you plan to work on the project).

GitHub configuration and uploading the project to the repository

Probably this and the following steps will not be possible during the meeting. Relax, it's already part of your project.

Creating a project on GitHub

1. First, have one member of your team create a project on *GitHub* similar to the configuration management lab (remember to add README.md and .gitignore for Java).
2. Edit the README.md file by providing a description of the project and additional information from the provided spreadsheet (https://help.github.com/articles/basic-writing-and-formatting-syntax/).
3. Grant permissions to team members and the teacher (Settings-> Collaborators).
4. Download the starter project (Architectural Frame) from Moodle and unpack it.
5. Edit the *pom.xml* file and adjust the project name, version etc.
6. Upload the contents of the unpacked directory to the repository (**not** the directory, but its contents so that *pom.xml* is in the main directory of the repository) using the "Upload files" option or Git commands.

Create Product Backlog and Sprint Backlog

The easiest solution for the project will be to use the *Issues* mechanism built into *GitHub*. The list of cases is available in the *Issues* tab of your *GitHub* instance. You can read more about this mechanism at: https://guides.github.com/features/issues/
We would like to leave some freedom and discretion to the team as to how exactly it will solve the problem of task management, paying attention only to a few possible solutions (the only non-negotiable requirement is to use *Issues* on *GitHub*). Dear Scrum Masters, this is a field for you to show off!

NOTE: The tools configuration and the introduction of the Product Backlog and Sprint Backlog must be made by the next meeting!!!

Perhaps the simplest approach would be to use Milestones to mark the range of Sprint Backlogs and Labels to distinguish tasks and requirements-items of the Product Backlog (it should be visible, e.g. labels "PB Item" or "Feature", "Bug" for Product Backlog items), adding information about the effort and importance of Product Backlog items or status. The article may be helpful: https://zube.io/blog/agile-project-management-workflow-for-github-issues/

You can also use the project mechanism that will allow you to create "Scrum / Kanban" boards with statuses: https://help.github.com/articles/about-project-boards/

Remember that you don't have to create the perfect solution from the very beginning. Scrum is an empirical approach and we hope that during the implementation of two Sprints and retrospectives at the end of each of them you will come up with a solution that will be satisfactory for you. What definitely needs to be achieved in this Sprint is to introduce the Product Backlog and Sprint Backlog for Sprint # 1 to the tool and actively use it.

GitHub Actions configuration

Configure the GitHub Actions tool to automatically build your project after any change and send email notifications in case of faulty builds. Also, use the lab materials for software building and continuous integration. Be sure to assign appropriate rights to team members and the teacher.